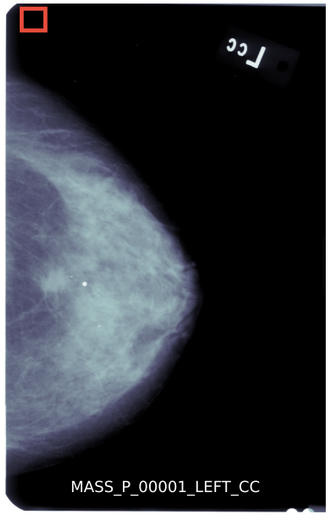
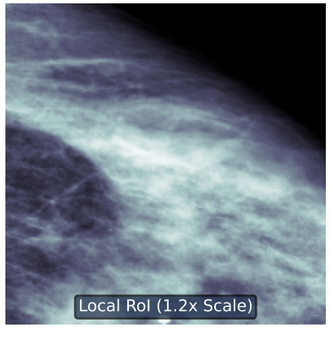
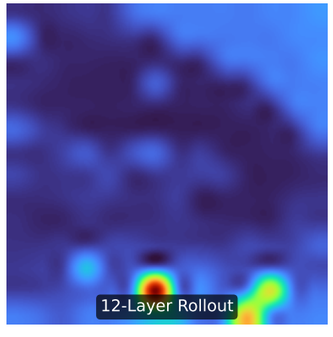
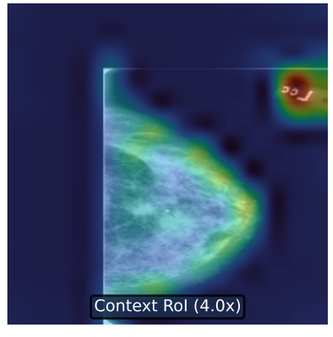
Case 1
Malignant












```
ATTENTION ROLLOUT FORMULATION
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1. Layer Attention Matrix:
   A^(l) = Softmax( Q^(l) K^(l)T / sqrt(d) )
   Aggregated across H=12 attention heads
   via mean reduction.

2. Residual Identity Accounting:
   A_hat^(l) = 0.5 * A^(l) + 0.5 * I
   Where identity matrix I represents the
   Transformer residual skip connection.

3. Recursive 12-Layer Rollout:
   R^(l) = A_hat^(l) * R^(l-1)
   Accumulates token interactions from
   Patch Embedding (Layer 0) through the
   fine-tuned Blocks (Layers 8-11).

4. Class Token Saliency Extraction:
   S_patch = R^(12)[CLS, 1:197]
   Interpolated to 224x224 input canvas.

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MANDATORY SCIENTIFIC DISCLAIMER:
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• Attention maps are descriptive representations
  of self-attention token weights.
• High saliency indicates feature correlation
  within the ViT receptive field, NOT clinical
  or pathological causation.
• AI-assisted research prototype only.
```

Case 2
Benign

